

Three Doors, Two Thirds, And Why It Stops At The Door

The most famous counterintuitive result in probability, and the most commonly misapplied. The switch is right, the reason is narrow, and that narrowness decides whether any of it reaches a trading screen. It does not. Knowing precisely why is worth more than the puzzle.

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01 The game

Three doors. One car. Two goats. You pick, the host opens a door, then offers a swap.

- You choose a door. It stays shut.
- **The host knows where the car is.** He opens one of the other two and **always shows a goat**. He is never allowed to show the car.
- The gut says fifty fifty because two doors remain. **That is the trap, and it is the whole lesson.**

02 The proof

Say you picked door one. The car is equally likely to be anywhere. That is the entire proof.

- **Car behind door 1.** The host may open 2 or 3, **his choice**. Staying wins.
- **Car behind door 2.** The host **must** open 3. Switching wins.
- **Car behind door 3.** The host **must** open 2. Switching wins.
- **Look at the middle column, not the doors.** In two rows of three the host is **forced**, and being forced is exactly when an action carries information. In the row he was free, **you had already won**.

THE ONE IDEA TO REMEMBER

The two thirds does not come from the doors. It comes from a host who **knows the answer and is obliged to reveal a goat**. Change what he knows and the maths changes, even though the view never does.

03 Questions

The five questions worth being able to answer out loud.

- **Why is it not fifty fifty with two doors left?** Your first pick was right one time in three and **nothing afterwards changed that**. The other door absorbs the remaining two thirds.

- **What if the host opens a door at random and it happens to be a goat? Then it genuinely is fifty fifty.** Same doors, same goat, same view, different odds. The result lives entirely in what the host knew.
- **Does it get clearer with more doors?** Pick 1 of 100, the host opens 98 goats. Your pick was **1 in 100**, the last door holds **99 in 100**. Three doors is the same trick, too small to feel.
- **Does flipping direction after a stop out raise your probability? No.** A stop out does not eliminate an outcome. It only resolves your own bet.
- **If a flip does not pay at 2 to 1, does it pay at 1 to 1? No, and it is harder.** A 2 to 1 structure breaks even at a **33.3 percent** hit rate. A 1 to 1 structure needs **50 percent**. The nearer target is easier to reach by exactly enough to cancel the smaller payout.

04 The catch

The puzzle transfers one idea to markets and blocks a second one. Both matter.

- **What transfers.** Ask of any print: **was that chosen, or was it forced?** A forced seller, a margin call or an index rebalance, says little about value and plenty about the next twenty minutes. A discretionary seller says the reverse. **Same print, different information.**
- **What does not transfer, and this is the important half.** The market is not the host. It does not know where the car is and **it owes you nothing**, so when the tape removes your alternatives, **no probability moves to what is left.**
- **A stop out eliminates nothing.** It resolves your bet and no other. **The outcomes are three, not two:** your side wins, the other wins, or **neither does**. Removing yours transfers probability to the union of the other two, never to the opposite alone. In chop both sides stop out and flipping pays the spread twice.
- **No risk to reward geometry manufactures an edge.** On a driftless walk the odds of reaching a target before a stop are **b divided by a plus b** (a is target distance, b is stop distance). Every shape has the same expectancy, which is zero, before costs.
- **After costs nothing cancels, and the tighter target is worse.** One ES contract costs about **5.74 dollars** a round trip. That is **1.4 percent** of a 400 dollar target and **2.9 percent** of a 200 dollar one. Same edge, twice the leak.

THE LINE TO KEEP

The edge was never in counting what is left. It was in knowing whether the thing revealed was chosen or forced, and the market is never obliged to show you anything.

SOURCES & NOTE

Sources: First posed by Steve Selvin in The American Statistician, 1975; Made famous by Marilyn vos Savant in Parade, 1990. Roughly 10,000 readers wrote in to say she was wrong, about 1,000 of them holding PhDs. She was right; Named for Monty Hall, host of the US game show Let's Make a Deal; Cost figures are Mann's own all in Tradovate rates, ES, August 2026. Questions four and five were asked by Shiv, August 4th 2026.

Educational, not financial advice. The maths is settled. What anyone does with a trading account is not.

Glossary

Monty Hall problem A three door puzzle where switching wins two times in three.

Prior What you believed before new evidence arrived. Your first door was one in three.

Base rate The underlying frequency of a thing before you condition on anything.

Conditional probability The odds of something given that something else is already known.

Forced action A move the actor had no choice about. This is where information hides.

Chop A market going nowhere. The third outcome the puzzle has no room for.

Random walk A path with no drift, where the next step is unrelated to the last.

Absorbing barrier A level that ends the game when touched. A stop and a target are both.

Expectancy Average result per trade, counting wins, losses and how often each happens.

Break even win rate How often you must be right for a structure to make zero. 2 to 1 needs 33.3 percent.

R multiple R Result measured in units of the risk taken. Losing one stop is minus one R.

Autocorrelation Whether a move predicts the next move. Near zero on short intraday horizons.
